

一、简答题 (40)

- 1) 在一个量子体系中, 有一个可能的状态 $\Psi, \Psi_2 = \Psi * \exp(ia)$
 Ψ_2 是这个体系的可能函数吗? 那么两者之间的关系是什么, 为什么?
- 2) 对于一个力学量 L , 有两个完备的态 Ψ_1 和 Ψ_2 , 写出一般的状态, 并写出体系可能值的概率和平均值, 依据是什么?
- 3) 对于 $\Psi(x,t)$, 有一组正交完备的基矢 Ψ_n , 请写出在这个基下的 $\Psi(x,t)$

$$12) \quad \Psi = C_1 \Psi_1 + C_2 \Psi_2 \quad \hat{L} \Psi_1 = L_1 \Psi_1, \quad \hat{L} \Psi_2 = L_2 \Psi_2$$

力学量的本征值正交归一且完备

体系可能值 L_1 概率 $|C_1|^2$

L_2 , $|C_2|^2$

$$\langle L \rangle = |C_1|^2 L_1 + |C_2|^2 L_2$$

$$13) \quad \Psi(x,t) = \sum_{n=1}^{\infty} C_n \Psi_n \cdot e^{-\frac{iE_n t}{\hbar}}$$

二、一个量子力学体系下, 有两个全同粒子, 有 n 个不同的态, 用 Ψ_n

表示, 他们的自旋波函数用 χ 表示, 请分别写出玻色子和费米子

情况的函数。

组合表示

$|11\rangle, |1-1\rangle, |10\rangle$ 满足交换对称性

$|00\rangle$ 满足交换反对称性

$$n_1 = n_2 = n, \quad \Psi_n = \Psi_n(x_1) \Psi_n(x_2) \left\{ \begin{array}{l} |11\rangle \\ |1-1\rangle \\ |10\rangle \end{array} \right. \quad \text{其中 } x_i \text{ 代表单粒子的全部坐标}$$

$$n_1 \neq n_2, \quad \Psi_{n_1 n_2} = \frac{1}{\sqrt{2}} [\Psi_{n_1}(x_1) \Psi_{n_2}(x_2) + \Psi_{n_2}(x_1) \Psi_{n_1}(x_2)] \left\{ \begin{array}{l} |11\rangle \\ |1-1\rangle \\ |10\rangle \end{array} \right.$$

$$\Psi_{n_1 n_2} = \frac{1}{\sqrt{2}} [\Psi_{n_1}(x_1) \Psi_{n_2}(x_2) - \Psi_{n_2}(x_1) \Psi_{n_1}(x_2)] |00\rangle$$

费米子

$$n_1 = n_2 = n, \quad \Psi_n = \Psi_n(x_1) \Psi_n(x_2) |00\rangle$$

$$n_1 \neq n_2, \quad \Psi_{n_1 n_2} = \frac{1}{\sqrt{2}} [\Psi_{n_1}(x_1) \Psi_{n_2}(x_2) + \Psi_{n_2}(x_1) \Psi_{n_1}(x_2)] |00\rangle$$

$$\Psi_{n_1 n_2} = \frac{1}{\sqrt{2}} [\Psi_{n_1}(x_1) \Psi_{n_2}(x_2) - \Psi_{n_2}(x_1) \Psi_{n_1}(x_2)] \left\{ \begin{array}{l} |11\rangle \\ |1-1\rangle \\ |10\rangle \end{array} \right.$$

三、一个态的波函数为 $\Psi = A(x + y + 2z)\exp(-\alpha r)$ 已给出 $l = 1$ 的球谐函数

- 1) 请问这个函数的 L^2 为多少
- 2) 求 L_z 的平均值
- 3) 求 L_z 为 $+h$ 的概率为多少
- 4) 求一个立体角 $d\Omega$ 内的概率为多少

$$\Psi = A(x + y + 2z) e^{-\alpha r}$$

$$x = r \sin\theta \cos\varphi$$

$$y = r \sin\theta \sin\varphi$$

$$z = r \cos\theta$$

$$\text{得 } \Psi = A r (\sin\theta \cos\varphi + \sin\theta \sin\varphi + 2\cos\theta) e^{-\alpha r}$$

$l=1$ 的球谐函数

$$Y_{11} = -\sqrt{\frac{3}{8\pi}} \sin\theta e^{i\varphi}$$

$$Y_{10} = \sqrt{\frac{3}{4\pi}} \cos\theta$$

$$Y_{1-1} = \sqrt{\frac{3}{8\pi}} \sin\theta e^{-i\varphi}$$

$$\cos\varphi = \frac{e^{i\varphi} + e^{-i\varphi}}{2}$$

$$\sin\varphi = \frac{e^{i\varphi} - e^{-i\varphi}}{2i}$$

$$\Psi = A r \left[\frac{1}{2} \frac{\sin\theta e^{i\varphi}}{-\sqrt{\frac{3}{8\pi}} Y_{11}} + \frac{1}{2} \frac{\sin\theta e^{-i\varphi}}{\sqrt{\frac{3}{8\pi}} Y_{1-1}} + \frac{1}{2i} \frac{\sin\theta e^{i\varphi}}{-\sqrt{\frac{3}{8\pi}} Y_{11}} - \frac{1}{2i} \frac{\sin\theta e^{-i\varphi}}{\sqrt{\frac{3}{8\pi}} Y_{1-1}} + 2 \frac{\cos\theta}{\sqrt{\frac{3}{4\pi}} Y_{10}} \right] e^{-\alpha r}$$

$$= A r e^{-\alpha r} \left(\sqrt{\frac{2\pi}{3}} Y_{1-1} - \sqrt{\frac{2\pi}{3}} Y_{11} + i \sqrt{\frac{2\pi}{3}} Y_{11} + i \sqrt{\frac{2\pi}{3}} Y_{1-1} + 4 \sqrt{\frac{\pi}{3}} Y_{10} \right)$$

$$= B r e^{-\alpha r} \left[(1+i) Y_{1-1} + (i-1) Y_{11} + 2\sqrt{2} Y_{10} \right]$$

$$\text{归一化 } \int B^2 r^2 e^{-2\alpha r} (2 Y_{1-1}^* Y_{1-1} + 2 Y_{11}^* Y_{11} + 8 Y_{10}^* Y_{10}) d\tau$$

$$= 12 B^2 \int_0^\infty r^4 e^{-2\alpha r} dr = 1$$

$$12 B^2 \cdot \frac{4!}{(2\alpha)^5} = 1$$

$$B^2 = \frac{\alpha^5}{9}$$

$$B = \frac{\sqrt{\alpha^5}}{3}$$

$$(1) \quad \hat{L}^2 = 2\hbar^2$$

$$(2) \quad \langle \hat{L}_z \rangle = 2B^2 \cdot (-\hbar) + 2B^2 \cdot \hbar = 0$$

$$(3) \quad \langle \hat{L}_x \rangle = 0, \quad \langle \hat{L}_y \rangle = 0$$

3) L_z 为 $+\hbar$ 的概率 $P = B^2 \cdot 2 = \frac{2A^2}{9}$

(4) $d\Omega = \frac{r^2 \sin\theta d\theta d\varphi}{r^2} = \sin\theta d\theta d\varphi$

$$P = \int \psi^* \psi d\Omega = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \psi^* \psi \sin\theta d\theta d\varphi$$

四、一个体系的哈密顿量为 $H = AS_y + BS_z$, A, B 为实数

1) 求这个体系的本征值和本征函数

2) 求 S_y

3) $S_y = +\hbar/2$ (guess) 在本征函数的概率

(1) $\hat{H} = A \cdot \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + B \cdot \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
 $= \frac{\hbar}{2} \begin{pmatrix} B & -iA \\ iA & -B \end{pmatrix}$

$$\hat{H} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\frac{\hbar}{2} \begin{pmatrix} B & -iA \\ iA & -B \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$- \left(\frac{\hbar}{2} B - E \right) \left(\frac{\hbar}{2} B + E \right) = \left(\frac{\hbar}{2} \right)^2 A^2$$

$$E^2 = \left(\frac{\hbar}{2} \right)^2 (A^2 + B^2)$$

$$E = \pm \frac{\hbar}{2} \sqrt{A^2 + B^2}$$

$$E_1 = \frac{\hbar}{2} \sqrt{A^2 + B^2} = \frac{\hbar}{2} \cdot C$$

$$\frac{\hbar}{2} \begin{pmatrix} B & -iA \\ iA & -B \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \frac{\hbar}{2} \cdot C \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\begin{cases} BC_1 - iAC_2 = C \cdot C_1 \\ C_1^2 + C_2^2 = 1 \end{cases}$$

$$C_2 = \frac{(B-C)C_1}{iA}$$

$$C_1^2 - \frac{(B-C)^2 C_1^2}{A^2} = 1$$

$$\frac{A^2 - B^2 - A^2 - B^2 + 2BC}{A^2} C_1^2 = 1$$

$$C_1 = \sqrt{\frac{A^2}{2BC - 2B^2}}$$

$$C_2 = \sqrt{\frac{2BC - 2B^2 - A^2}{2BC - 2B^2}}$$

$$|+\rangle = \frac{1}{\sqrt{2}}$$

五、对于一个在 xy 平面运动的转子，他的哈密顿量为 $H = \frac{L_z^2}{2I}$

1) 求能量本征值和本征函数

2) 在一个微扰 $H' = A \sin(2\varphi)$ 下，求基态和第一激发态的一阶微扰和零级波函数

1) 有征值 $\frac{m^2 \hbar^2}{2I}$
有征函数 $\psi(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi}$

$$\hat{H}\psi = E\psi$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi}$$

$$-\frac{\hbar^2}{2I} \frac{d^2}{d\varphi^2} \psi = E\psi$$

$$\frac{d^2 \psi}{d\varphi^2} = -\frac{2IE}{\hbar^2} \psi$$

$$= -k^2 \psi$$

2) 第一激发态 $m = \pm 1$ ，二度简并， $E_1 = \frac{\hbar^2}{2I}$

$$m=1, \psi = \frac{1}{\sqrt{2\pi}} e^{i\varphi} = \varphi_1$$

$$m=-1, \psi = \frac{1}{\sqrt{2\pi}} e^{-i\varphi} = \varphi_2$$

$$\psi^{(0)} = C_1 \varphi_1 + C_2 \varphi_2$$

$$\begin{pmatrix} H'_{11} & H'_{12} \\ H'_{21} & H'_{22} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E^{(1)} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$H'_{11} = 0 \quad H'_{22} = 0$$

$$H'_{12} = -\frac{A}{2i} \quad H'_{21} = \frac{A}{2i}$$

$$\begin{pmatrix} 0 & -\frac{A}{2i} \\ \frac{A}{2i} & 0 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E^{(1)} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\text{解久期方程 } (E^{(1)})^2 = \frac{A^2}{4}$$

$$E^{(1)} = \pm \frac{A}{2}$$

$$E^{(1)} = \frac{A}{2}, \psi_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix}$$

$$E^{(1)} = -\frac{A}{2}, \psi_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix}$$

修正: $E = \frac{\hbar^2}{2I} + \frac{A}{2}, \psi^{(0)} = \frac{1}{\sqrt{2}} (i\psi_1 + \psi_2)$

$$E = \frac{\hbar^2}{2I} - \frac{A}{2}, \psi^{(0)} = \frac{1}{\sqrt{2}} (-i\psi_1 + \psi_2)$$

12) $H' = A \sin 2\varphi$

① 基态 $m=0$ 无简并

$$E_0 = E_0^{(0)} + E_0^{(1)}$$

$$E_0^{(1)} = H'_{00} = \langle \psi_0 | H' | \psi_0 \rangle$$

$$= \int_0^{2\pi} \frac{A}{2\pi} \cdot \sin 2\varphi d\varphi$$

$$= 0$$

$$H'_{mn} = \int \psi_m^* A \sin 2\varphi \psi_n d\tau$$

$$= \frac{A}{2\pi} \int e^{i(n-m)\varphi} \sin 2\varphi d\tau$$

$$= \frac{A}{2\pi} \int_0^{2\pi} e^{i(n-m)\varphi} \cdot \frac{e^{2i\varphi} - e^{-2i\varphi}}{2i} d\varphi$$

$$= \frac{A}{4\pi i} \int_0^{2\pi} [e^{i(n-m+2)\varphi} - e^{i(n-m-2)\varphi}] d\varphi$$

$$= \frac{A}{2i} [\delta_{n,m+2} - \delta_{n,m-2}]$$

$$\psi_0^{(1)} = \frac{H'_{20}}{E_0^{(0)} - E_2^{(0)}} \psi_2^{(0)} + \frac{H'_{-20}}{E_0^{(0)} - E_{-2}^{(0)}} \psi_{-2}^{(0)}$$

$$= \frac{\frac{A}{2i}}{0 - \frac{4\hbar^2}{2I}} \cdot \frac{1}{\sqrt{2}} e^{2i\varphi} + \frac{\frac{-A}{2i}}{0 - \frac{4\hbar^2}{2I}} \cdot \frac{1}{\sqrt{2}} e^{-2i\varphi}$$

$$= -\frac{AI}{4\hbar^2 \sqrt{2}} e^{2i\varphi} + \frac{AI}{4\hbar^2 \sqrt{2}} e^{-2i\varphi}$$

六、一个体系的哈密顿量为 $H = \frac{eB}{2m} S_x$, $t=0$ 时, S_z 为正, 求 $t>0$ 时

1) 体系波函数

2) 求任意时刻, S_y 为 $+\frac{\hbar}{2}$ 或 $-\frac{\hbar}{2}$ 的概率

$$1) \hat{H} = \frac{eB\hbar}{4m} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad E_1 = \frac{eB\hbar}{4m} = \omega \hbar |\psi_+\rangle = \frac{1}{2} \hbar \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$E_2 = -\frac{eB\hbar}{4m} = -\omega \hbar |\psi_-\rangle = \frac{1}{2} \hbar \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{aligned} |\psi\rangle &= C_1 e^{-\frac{iE_1 t}{\hbar}} |\psi_+\rangle + C_2 e^{-\frac{iE_2 t}{\hbar}} |\psi_-\rangle \\ &= \frac{C_1}{\hbar} e^{-i\omega t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{C_2}{\hbar} e^{i\omega t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \frac{1}{\hbar} \begin{pmatrix} C_1 e^{-i\omega t} + C_2 e^{i\omega t} \\ C_1 e^{-i\omega t} - C_2 e^{i\omega t} \end{pmatrix} \end{aligned}$$

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\hbar} \begin{pmatrix} C_1 + C_2 \\ C_1 - C_2 \end{pmatrix}$$

$$\begin{aligned} C_1 + C_2 &= \hbar \\ C_1 - C_2 &= 0 \end{aligned} \quad C_1 = C_2 = \frac{1}{2}\hbar$$

$$\therefore |\psi(t)\rangle = \begin{pmatrix} \cos \omega t \\ -i \sin \omega t \end{pmatrix}$$

$$(2) \quad S_y = \frac{\hbar}{2}, \quad |\psi_1\rangle = \frac{1}{\hbar} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$S_y = -\frac{\hbar}{2}, \quad |\psi_2\rangle = \frac{1}{\hbar} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$P_{S_y = \frac{\hbar}{2}} = |\langle \psi_1 | \psi(t) \rangle|^2 = \frac{1}{2} \left| \begin{pmatrix} 1 & i \end{pmatrix} \begin{pmatrix} \cos \omega t \\ -i \sin \omega t \end{pmatrix} \right|^2$$

$$= \frac{1}{2} (\cos \omega t - \sin \omega t)^2$$

$$\begin{aligned} P_{S_y = -\frac{\hbar}{2}} &= |\langle \psi_2 | \psi(t) \rangle|^2 = \frac{1}{2} \left| \begin{pmatrix} 1 & -i \end{pmatrix} \begin{pmatrix} \cos \omega t \\ -i \sin \omega t \end{pmatrix} \right|^2 \\ &= \frac{1}{2} (\cos \omega t + \sin \omega t)^2 \end{aligned}$$